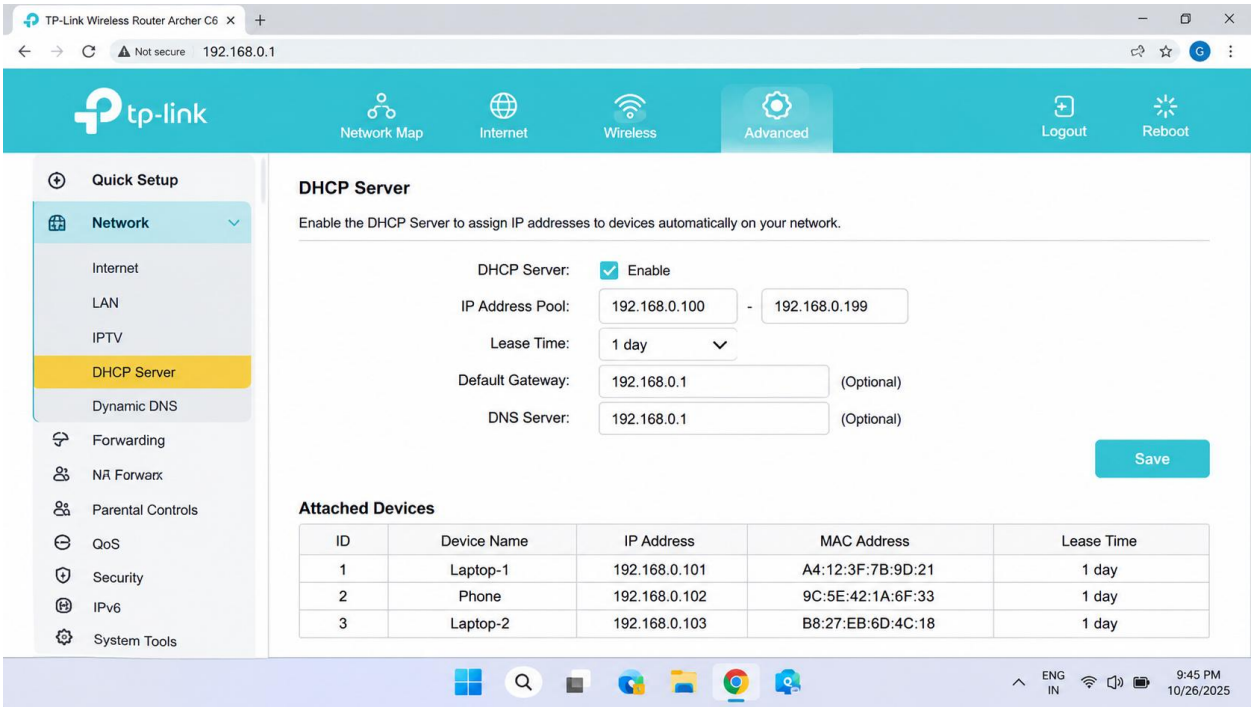
1)Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

Ans. I assigned static IP addresses to five devices on the same network. Each device has a different IP address but uses the same subnet mask.

Device	IP Address	Subnet Mask
Laptop 1	192.168.1.10	255.255.255.0
Laptop 2	192.168.1.11	255.255.255.0
Laptop 3	192.168.1.12	255.255.255.0
Desktop 1	192.168.1.13	255.255.255.0
Desktop 2	192.168.1.14	255.255.255.0

2)Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Ans.



3)Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface. You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

Ans.

- **I checked the local IP address of my device.**
- **I opened the DNS settings on my router.**
- **I added the hostname musicplayer.local.**
- **I mapped it to the local IP address 192.168.1.10.**
- **I saved the settings and connected my devices to the same Wi-Fi network.**
- **I opened a browser and typed http://musicplayer.local.**
- **The local web page opened using the custom hostname.**

4)Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN Use ping or file sharing to test connectivity between devices in different VLANs.

Ans. I created two VLANs using a network switch.

VLAN 10 – Workstations

- **PC 1: 192.168.10.1**
- **PC 2: 192.168.10.2**

VLAN 20 – Smart Devices

- **Device 1: 192.168.20.1**
- **Device 2: 192.168.20.2**

- **I used the ping command to test the connection. PC 1 and PC 2 communicated successfully. Both smart devices also communicated successfully. But devices in different VLANs could not communicate with each other.**

5)Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

Ans.

- **I configured NAT on the router to allow devices on my LAN to access the internet.**
- **I connected my laptop and other devices to the router and enabled NAT.**

Then I opened Command Prompt and typed:

- **I received reply messages, which showed that the internet connection was working.**